VISHNU CHOUNDUR

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SUMMARY

Data and Analytics Engineer with 5+ years of extensive experience translating business and operational needs into scalable, data-driven analytical solutions that support enterprise objectives and measurable outcomes. Proven expertise in building robust data integration and ETL architectures processing millions of records daily using incremental and batch strategies, while converting complex normalized source systems with varying timestamps into denormalized analytical datasets for reporting and decision-making. Strong hands-on experience applying SCD Type 1 and Type 2 transformations, advanced SQL, and stored procedures to meet evolving business requirements and ensure accurate historical and current-state analysis. Skilled in designing governed analytical data models and frameworks that enable consistent performance tracking and optimization. Extensive experience working with Snowflake, SQL Server, SSMS, PostgreSQL, Python, and SQL across cloud and on-prem environments, along with orchestration and streaming tools such as Airflow and Kafka. Adept at data monitoring, data quality validation, and governance practices to ensure analytical reliability, while collaborating with stakeholders to deliver trusted insights and promote data-driven decision-making at scale using cloud-based platforms

TECHNICAL SKILLS

ETL / Data Integration & Orchestration	Airflow, dbt, Informatica PowerCenter, Talend, StreamSets, SSIS, ETL/ELT Frameworks, PySpark, Python, Workflow Scheduling, CI/CD for Data Pipelines
Operating Systems	Linux (RHEL), Windows Server
Big Data & Cloud Platforms	AWS (S3, EMR, Redshift, DynamoDB, SQS, Lambda, EC2), Azure (Data Lake, Cosmos DB, Azure Databricks, Azure Data Factory, Azure Functions, Logic apps), GCP (BigQuery, Cloud Storage)
Databases & Data Modeling	Snowflake, SQL Server, PostgreSQL, Oracle, Teradata, Stored Procedures, Window Functions, Star & Snowflake Schema, ER Modeling, SCD Type 1 & 2, Incremental & Batch Processing
Scripting & Automation	Python, SQL, T-SQL, PL/SQL, Bash, Shell Scripting, JSON, YAML, SQL Optimization, Automation of Data Pipelines
Data Governance & Quality	Data Governance Frameworks, Metadata Management, Lineage Tracking, Data Quality Validation, Monitoring & Alerting, Compliance Controls, Performance Optimization
Analytics, BI & Reporting	Power BI, Tableau, Looker Studio, KPI Design, Dashboard Development, Trend Analysis, BAU Reporting, Analytical Enablement
DevOps/Monitoring tools	Jenkins, GitLab CI, DevOps, GitHub, Terraform, Git, SVN, Splunk, Grafana
Core Engineering Competencies	Data Engineering at Scale, Normalized→Denormalized Transformations, Complex Timestamp Alignment, Data Pipeline Reliability, Workflow Orchestration, CI/CD, Agile Delivery
Professional Skills	Analytical Thinking, Problem Solving, Communication with Stakeholders, Requirement Translation, Cross-Functional Collaboration, Prioritization, Multi-Project Ownership

PROFESSIONAL EXPERIENCE

Data Engineer | Morgan Stanley | USA ***Jun 2024 – Present***

- Contributed to the large-scale data migration from Snowflake to Azure Databricks, transiting of raw data from ADLS Gen2 and building ETL pipelines using PySpark, enhancing data processing speed and reliability by 35%.
- Utilized Azure Data Factory and Apache Airflow for end-to-end workflow management, implementing automated DAGs to streamline ETL processes and troubleshoot with detailed log analysis.
- Integrated Azure Event Grid with Snow Pipe for real-time data ingestion into Snowflake’s staging area, reducing latency by 50% and enhancing data availability across Power BI-based data dashboards.
- Refactored and optimized SQL transformations using DBT, translating existing Snowflake scripts into modular, maintainable SQL code within Azure Databricks and improving query performance with Delta Lake optimizations by 30%.
- Developed custom Python scripts using pandas and NumPy for rigorous data validation and reconciliation post-migration, ensuring accuracy and consistency across Azure Databricks and Snowflake environments.
- Leveraged dynamic pipeline generation capabilities in Azure Data Factory and Airflow to support seamless data migration across development, staging, and production environments, facilitating data synchronization and process automation.
- Handled diverse data formats (Parquet, Avro, JSON, CSV) within Azure Databricks, using PySpark to efficiently process and transform data, optimizing both storage and query performance.
- Utilized Azure Event Hubs for real-time data streaming within the migration framework, automating processing workflows with Azure Functions and Logic Apps to enhance data accessibility and integration.

- Automated infrastructure deployment and configuration using Terraform (Iac) and Azure Resource Manager (ARM) templates, ensuring environment consistency, reducing setup time by 60% and streamlining the setup of data engineering resources.
- Managed system administration and data pipeline orchestration tasks using PowerShell on Windows and Bash on Unix/Linux, automating server management, log monitoring, and scheduling ETL processes for seamless data operations.

Environment: Azure Databricks, Azure Data Lake Storage (ADLS), Snowflake, Azure Event Grid, Azure Functions, Logic Apps.

Data Engineer | Accenture | India **May 2018 – Jul 2022**

Project 1: Health Care **Aug 2020 – Jul 2022**

- Designed and developed end-to-end data pipelines by implementing a healthcare data Lakehouse architecture on Azure, leveraging ADLS for scalable data storage, Azure Data Factory (ADF) for ETL processing, and Azure Synapse Analytics for data warehousing, ensuring efficient processing and achieving a 25% reduction in operational storage cost.
- Standardized and enriched healthcare datasets (EMR, claims datasets) using a multi-layered approach (Bronze, Silver, Gold) to ensure clean, accurate, and analytics-ready data for diverse business use cases.
- Engineered complex data transformation workflows using PySpark, processing EMR and Claims datasets to perform data cleaning, enrichment, and aggregation; implemented optimized join strategies, partitioning, and caching techniques.
- Implemented Slowly Changing Dimensions (SCD) Type 1 and Type 2 in Azure Synapse Analytics for accurate historical data tracking and consistent reporting.
- Developed interactive Power BI dashboards, delivering insights into key performance indicators (KPIs) such as revenue trends, payment timelines, and accounts receivable metrics, helping stakeholders identify a 15% improvement in payment timelines.
- Developed and maintained CI/CD pipelines using Azure DevOps and Jenkins for automated build, testing, and deployment, ensuring seamless integration and delivery of new data solutions.
- Implemented predictive analytics by integrating Azure Machine Learning, TensorFlow, and PyTorch on Azure HDInsight; applied Scikit-learn and NumPy for clustering, decision trees, and testing, optimizing scheduling and route efficiency.
- Monitored data pipeline performance using Azure Monitor, Splunk, and Prometheus, setting up alerts and creating custom dashboards to track KPIs related to ETL processes, error rates, and data latency.

Environment***:*** Azure Data Lake Storage, Azure Synapse Analytics, Azure HDInsight, Azure Machine Learning

Project 2: Telecom Data Integration & Analytics Platform **May 2018 – Jul 2020**

- Built and optimized a data integration and analytics solution utilizing cloud platforms and advanced data engineering practices, enhancing data accessibility, real-time processing, and performance for enterprise-wide data insights.
- Developed and maintained dynamic Tableau dashboards across 15 operational sites, resulting in a 20% improvement in decision-making and a 40% reduction in query times by leveraging optimized live data connections and interactive filtering.
- Implemented comprehensive data profiling and quality assurance processes to cleanse, match, and consolidate data, ensuring high data integrity and preparing datasets for warehouse loading.
- Designed Star Schema and Snowflake models for OLAP, applying data mapping techniques and analytical insights to support complex business challenges and enable efficient reporting.
- Developed interactive dashboards in Tableau and Power BI for monitoring claims metrics and fraud alerts, driving a 25% increase in 25% data-driven decision-making through effective data visualization and tailored reporting.
- Automated user management on Tableau Server with a Python script leveraging Tableau Server API, removing 3,400 inactive users and 12 unused sites, which enhanced server security and optimized resource management by 80%.
- Collaborated with business stakeholders to gather requirements, translate them into actionable technical specifications, and document data processes, ensuring accuracy and alignment throughout project stages.
- Handled data mapping and transformation workflows to ensure seamless integration and consistency across multiple data sources, leveraging ETL tools and custom SQL procedures for accurate data flow.
- Implemented data visualization best practices in Tableau, including drill-down capabilities, custom calculations, and storytelling techniques to optimize dashboard usability and stakeholder engagement.

Environment:. Informatica, Talend, custom SQL procedures, Python automation scripts, SQL (CTEs, Windows Functions).

EDUCATION

- **The University of Memphis, Memphis, Tennessee** **Aug 2022 - Dec 2024**

  Master of Science, **Data Science | GPA:** 3.68

  Course work: Database Systems, Data Mining, Machine Learning, Advanced Statistical Learning, Biostatistics, NLP